EET202

HINES & TRANSFORMERS

MODULE:2

:5 – Generator Characteristics

e performance characteristics of DC generators

Prepared By,

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EEE Department
SCET Kodakara

haracteristics of DC Generators

acteristics (Open Circuit Characteristics/ No ation Curve)

en **generated voltage and field current** keeping

ristics

racteristics

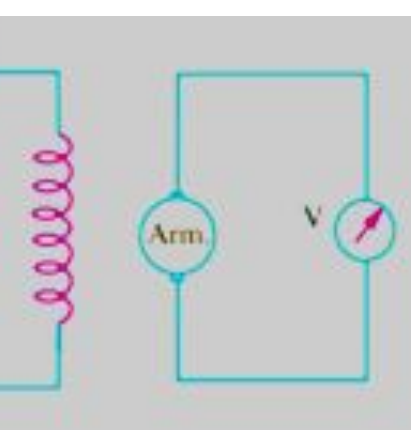
ween **terminal voltage and load current** keeping and speed constant.

racteristics

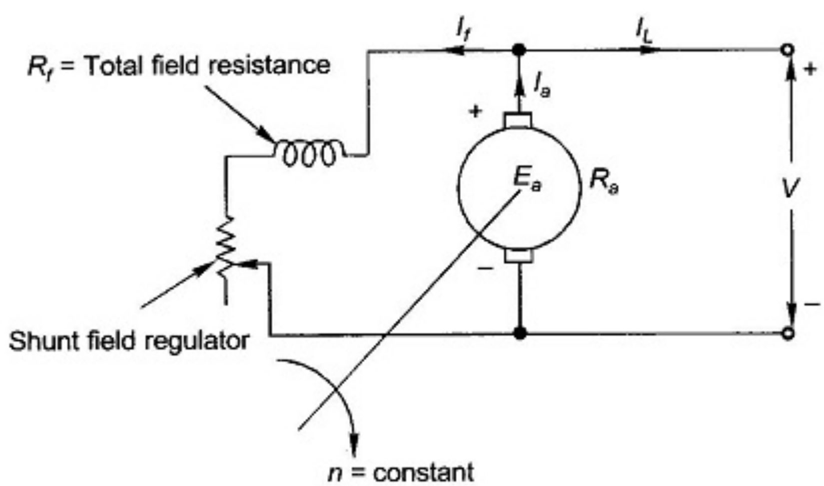
ween generated voltage and load current keeping and speed constant.

en Circuit Characteristics (OCC)

no load magnetisation curve.
which shows the relation between generated
d field current (I_f) keeping the speed (N)

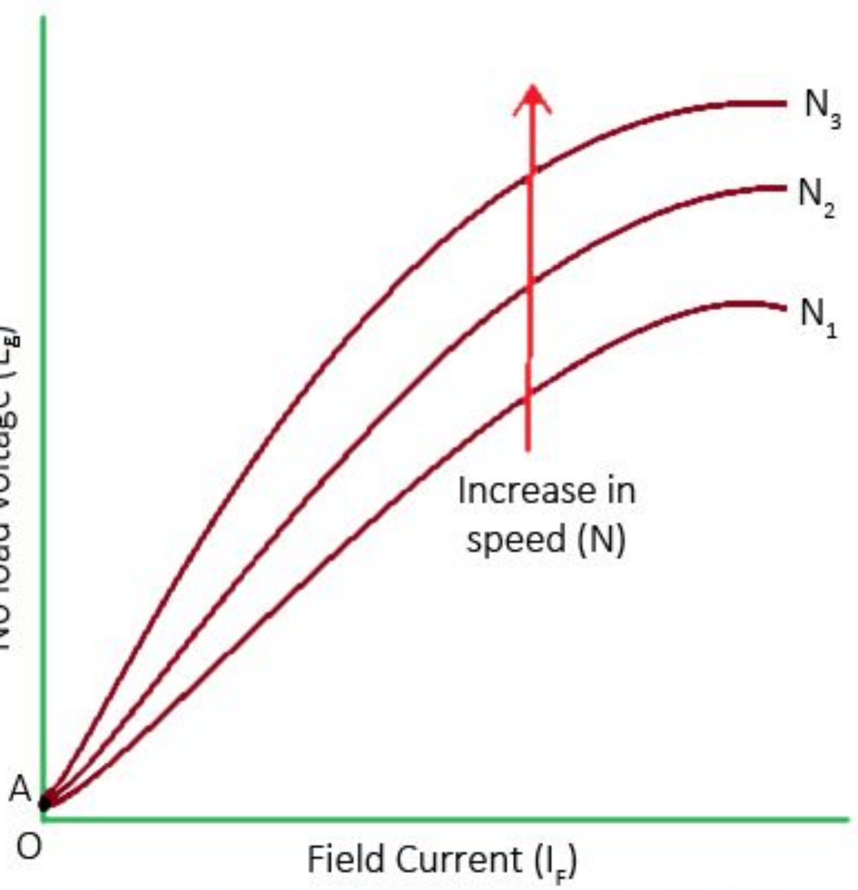


ted Generator



Self Excited Generator

OCC of DC Generator

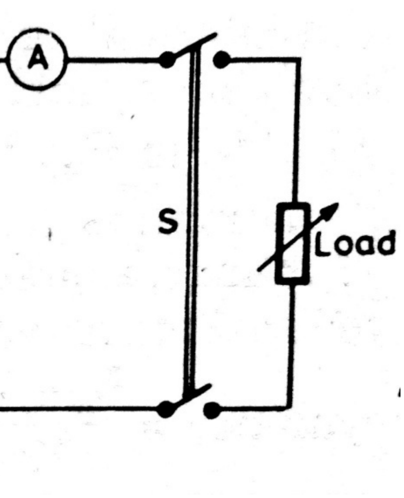


Open Circuit Characteristic (O.C.C.)

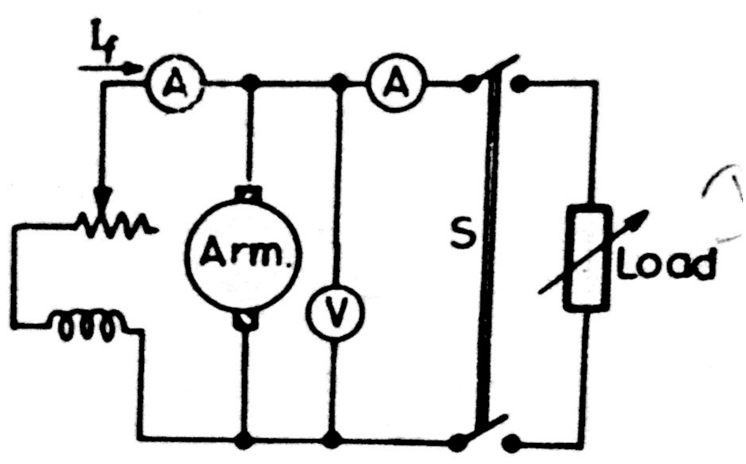
oad Characteristics

load magnetisation curve.

which shows the relation between terminal d field current (I_f) keeping the speed (N) and ent (I_a) constant.

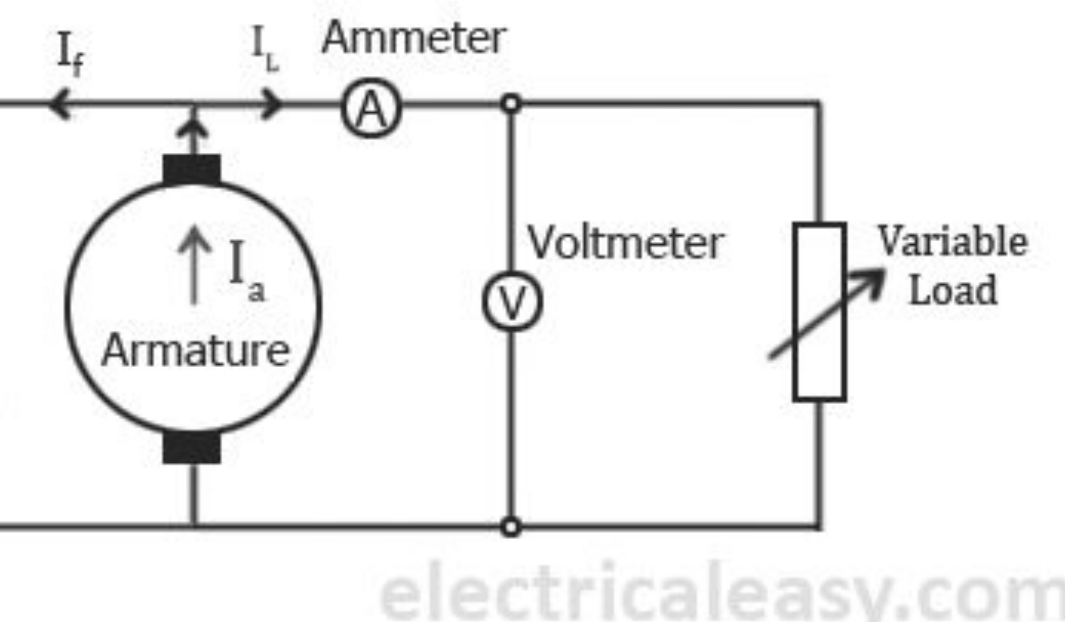


ted Generator



Self Excited Generator

l & Internal Characteristics



al Characteristics for Separately Excited Generator

t rated speed and its field winding is excited to give
age at no load.
and change the load current in steps and note the
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voltage
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circuit

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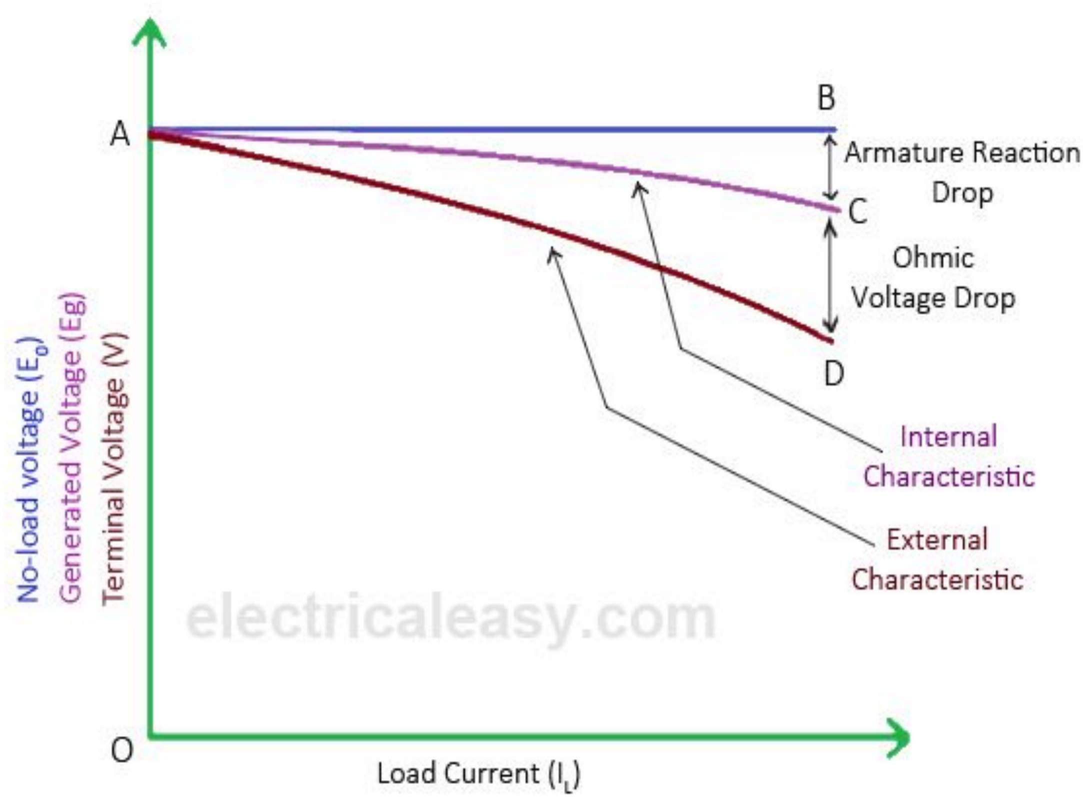

nal Characteristics for Shunt Generator

t rated speed and its field winding is excited to give
age at no load.
and change the load current in steps and note the
rresponding to different loads.
inal voltage with increase in load is due to the
ce drop, reduction in main flux due to armature
rther reduction in field current.

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and field

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creases and
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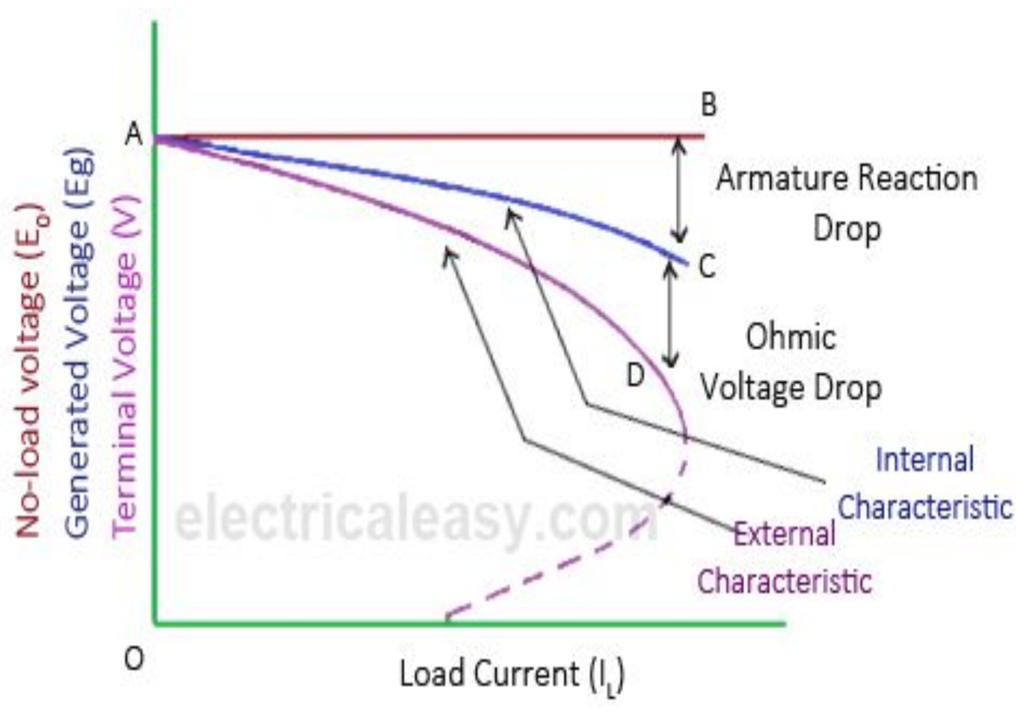

nal Characteristics for Series Generator

and change the load current in steps and note the
rresponding to different loads.

al voltage with increase in load is due to the voltage
armature reaction and armature and field winding